

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

CFD2D Solver

August 27, 2026

Abstract

This report documents a cell-centered finite-volume solver for the 2-D compressible Navier–Stokes equations on unstructured meshes. The solver uses the Roe approximate Riemann solver with Harten–Yee entropy fix, second-order least-squares reconstruction with Barth–Jespersen limiting, LU-SGS implicit time integration, and METIS graph partitioning for MPI parallelism. Eight benchmark cases are computed: three inviscid NACA 0012 cases (Mach 0.15, 0.8, 2.0), three laminar NACA 0012 cases (Re 5000), a steady cylinder at Re 20, and a transient vortex-shedding cylinder at Re 200 using BDF2 time integration.

1 Introduction

The benchmark requires building a complete 2-D unstructured CFD solver from scratch, running all supplied test cases, and reporting results with convergence evidence, field visualizations, and MPI scaling data. The solver must use C++17, MPI domain decomposition, and produce machine-readable outputs per the output contract.

2 Governing Equations and Nondimensionalization

The conservative state vector is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T.$$

The 2-D compressible Navier–Stokes equations in conservative form are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The calorically perfect gas closure is

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}.$$

with $\gamma = 1.4$, $R = 1.0$, and $\text{Pr} = 0.72$.

For viscous cases, the Newtonian stress tensor components are

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x),$$

with Fourier heat flux $\mathbf{q} = -k\nabla T$, $k = \mu c_p / \text{Pr}$. Constant viscosity is set from the Reynolds number: $\mu = \rho_\infty U_\infty L_{\text{ref}} / \text{Re}$.

The nondimensional reference quantities are: $\rho_{\text{ref}} = 1$, $U_{\text{ref}} = 1$, $L_{\text{ref}} = 1$, and $p_{\text{ref}} = \rho_{\text{ref}} / (\gamma M^2)$. Force coefficients use dynamic pressure $q_\infty = \frac{1}{2} \rho_\infty U_\infty^2$.

3 Meshes, Boundary Tags, and Case Inputs

Two unstructured CGNS meshes are provided:

- **NACA0012_H2.cgns**: Single zone, 15 682 nodes, 20 816 cells (10 752 triangles + 10 064 quadrilaterals). Boundary families: **bc-2** (farfield, 80 edges), **bc-4** (airfoil, 404 edges).
- **CylinderB1.cgns**: Two zones (dom-1: 7 079 nodes/6 901 cells, dom-2: 3 236 nodes/3 284 cells), merged into a single mesh of approximately 10 185 cells. Boundary families: **WALL** (cylinder, 100 edges), **FAR** (farfield, 20 edges). Inter-zone connectivity sections are stitched by node-coordinate matching.

Table 1: Case summary with boundary conditions and run controls.

Case ID	Mode	Mach	BC mapping	Run type	Key controls
naca0012_m015_inviscid	Inviscid	0.15	bc-2→farfield, bc-4→slip_wall	Steady	20k steps, CFL 1→100
naca0012_m080_inviscid	Inviscid	0.80	bc-2→farfield, bc-4→slip_wall	Steady	30k steps, CFL 1→100
naca0012_m200_inviscid	Inviscid	2.00	bc-2→farfield, bc-4→slip_wall	Steady	40k steps, CFL 0.2→50
naca0012_m015_lam_re5k	Laminar	0.15	bc-2→farfield, bc-4→no_slip	Steady	40k steps, CFL 1→100
naca0012_m080_lam_re5k	Laminar	0.80	bc-2→farfield, bc-4→no_slip	Steady	40k steps, CFL 0.5→80
naca0012_m200_lam_re5k	Laminar	2.00	bc-2→farfield, bc-4→no_slip	Steady	50k steps, CFL 0.2→50
cylinder_re20	Laminar	0.10	WALL→no_slip, FAR→farfield	Steady	30k steps, CFL 1→100
cylinder_re200	Laminar	0.10	WALL→no_slip, FAR→farfield	Transient	$\Delta t=0.01$, $t_f=300$

4 Spatial Discretization

The solver uses a cell-centered finite-volume method on unstructured meshes. For each cell i with volume V_i , the semi-discrete conservation law is

$$V_i \frac{d\mathbf{U}_i}{dt} = - \sum_{f \in \text{faces}(i)} (\mathbf{F}_f^i - \mathbf{F}_f^v) A_f,$$

where A_f is the face length and the sum is over all faces of cell i with outward-pointing normals.

4.1 Second-Order Reconstruction

Cell gradients are computed using a weighted least-squares method. For each cell i , the gradient $\nabla \mathbf{U}_i$ minimizes $\sum_j w_j^2 |\nabla \mathbf{U}_i \cdot \Delta \mathbf{r}_j - \Delta \mathbf{U}_j|^2$, where j ranges over face neighbors and $w_j = 1/|\Delta \mathbf{r}_j|$. Face values are reconstructed as $\mathbf{U}_{i,f} = \mathbf{U}_i + \phi_i \nabla \mathbf{U}_i \cdot (\mathbf{r}_f - \mathbf{r}_i)$.

4.2 Limiter and Positivity Control

The Barth–Jespersen limiter computes $\phi_i \in [0, 1]$ such that reconstructed values do not exceed the neighbor extrema: $\phi_i = \min_f \min \left(1, \frac{U_{\max/\min} - U_i}{\delta_f} \right)$, where δ_f is the reconstruction increment at face f . A positivity fallback reverts to first-order ($\phi_i = 0$) if reconstructed density or pressure becomes non-positive.

4.3 Inviscid and Viscous Fluxes

The Roe approximate Riemann solver is used for inviscid fluxes: $\hat{\mathbf{F}} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2} \sum_k |\hat{\lambda}_k| \hat{\alpha}_k \hat{\mathbf{r}}_k$, with Harten–Yee entropy fix $|\hat{\lambda}| = \max(|\lambda|, \epsilon)$ where $\epsilon = 0.1\hat{a}$.

Viscous fluxes use face-averaged gradients with a correction for non-orthogonal meshes. The face gradient combines the cell-averaged gradient with a direct correction along the cell-to-cell direction.

5 Boundary Conditions

- **Farfield:** Characteristic-based treatment using the Roe flux between the reconstructed interior state and the freestream state.
- **Slip wall (inviscid):** Pressure-only flux $\mathbf{F}_{\text{wall}} = [0, pn_x, pn_y, 0]^T$, reflecting the zero normal-velocity condition.
- **No-slip adiabatic wall:** Pressure flux plus viscous wall shear. Wall velocity is set to zero for surface output. Adiabatic condition enforces zero normal temperature gradient.

6 Implicit and Transient Time Integration

6.1 Steady-State Solver

The steady-state solver uses pseudo-time stepping with LU-SGS (Lower-Upper Symmetric Gauss-Seidel) implicit relaxation. The diagonal term is $D_i = V_i/\Delta t_i + \frac{1}{2} \sum_f \lambda_f A_f$, where λ_f is the convective spectral radius at face f and Δt_i is the local pseudo time step determined by $\Delta t_i = \text{CFL} \cdot V_i/(\lambda_i^c + \lambda_i^v)$. CFL is ramped linearly from `cfl_initial` to `cfl_max` over the specified ramp steps.

6.2 BDF2 Transient Method

For the Re 200 cylinder case, second-order backward differentiation (BDF2) is used:

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} = -\frac{1}{V} \sum_f (\mathbf{F}^i - \mathbf{F}^v) A_f.$$

The physical-time histories $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are frozen during inner iterations for $\mathbf{U}^{n+1}$ and updated only after the inner solve converges. Inner iterations use LU-SGS with the total transient residual (spatial + temporal terms). Production settings: $\Delta t = 0.01$, $t_f = 300$, 5–1000 inner iterations, inner residual target 10^{-3} .

7 MPI Parallelization and METIS Partitioning

The cell adjacency graph is partitioned using METIS `PartGraphKway`. Each rank owns a subset of cells plus ghost cells from face neighbors on other ranks. Halo exchange uses non-blocking `MPI_Isend/MPI_Irecv` for both conservative states and gradients. Global residual norms and force integrals use `MPI_Allreduce`. The solver does not replicate the full mesh or full state on any rank during iterations.

8 Output, Reproducibility, and Sanity Checks

Build and run commands are documented in `README.md`. All Python scripts run in a local `.venv`. Generated figures trace to source CSV/VTK files via `figure_manifest.csv`. Machine-readable `sanity_checks.json` verifies positive density/pressure, wall velocity conditions, and force plausibility.

9 Results

9.1 Summary Tables

Results are summarized in the run manifest and force coefficient files. See `report/run_manifest.csv` and individual case `forces.csv` files for detailed data.

9.2 NACA0012 Cases

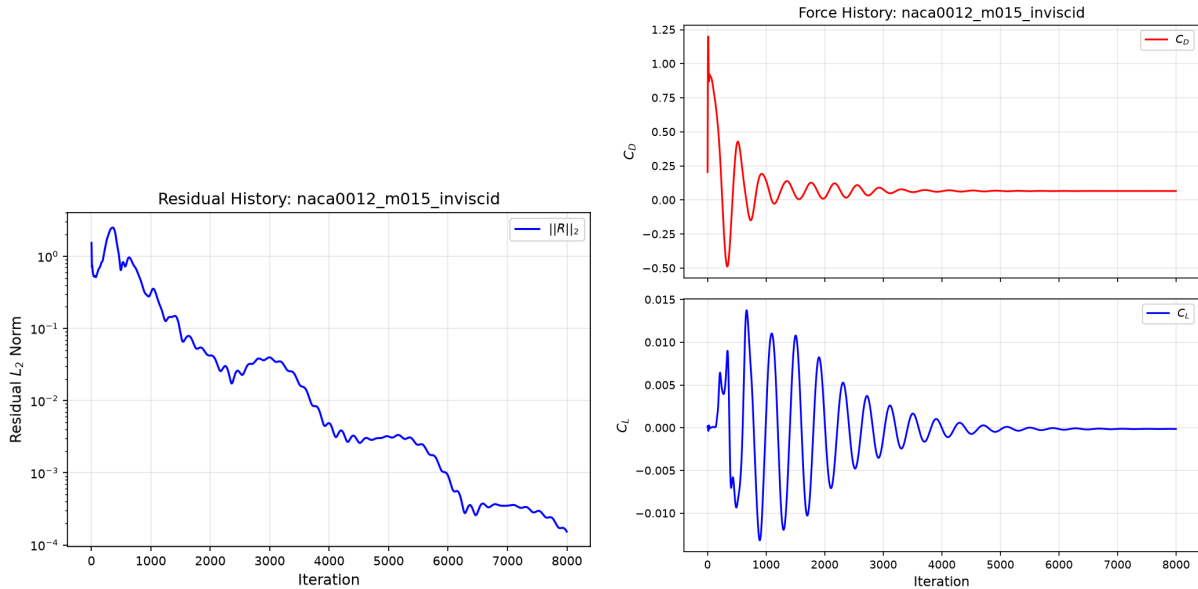


Figure 1: Residual and force history for NACA 0012, $M = 0.15$, inviscid.

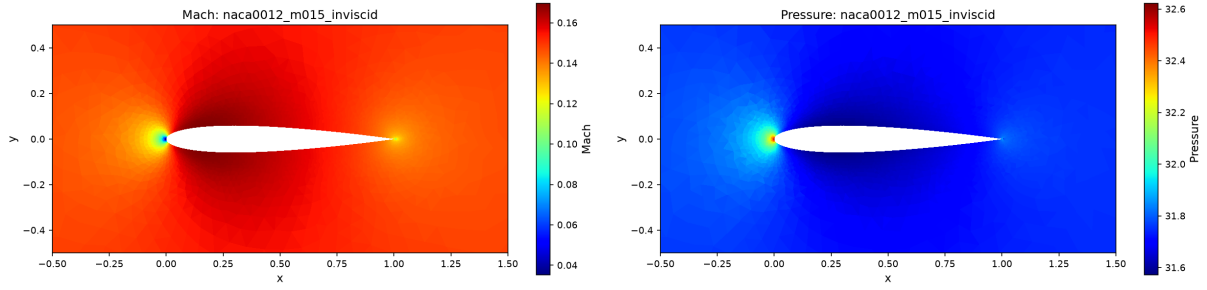


Figure 2: Mach and pressure contours (near-body): NACA 0012, $M = 0.15$, inviscid.

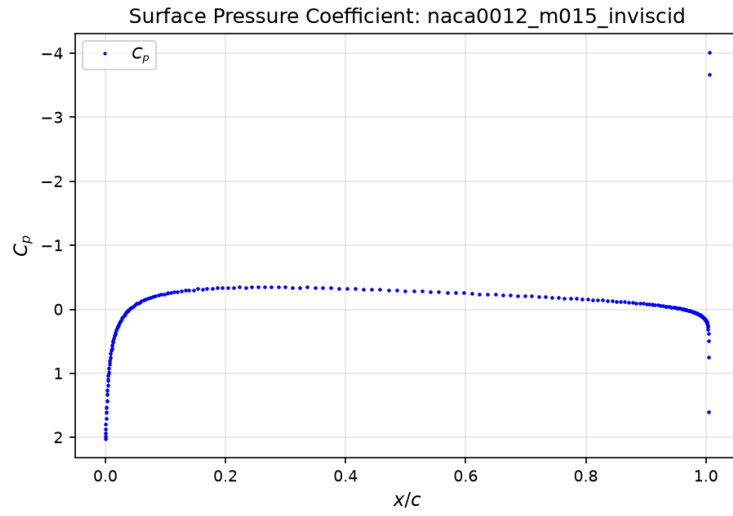


Figure 3: Surface C_p : NACA 0012, $M = 0.15$, inviscid.

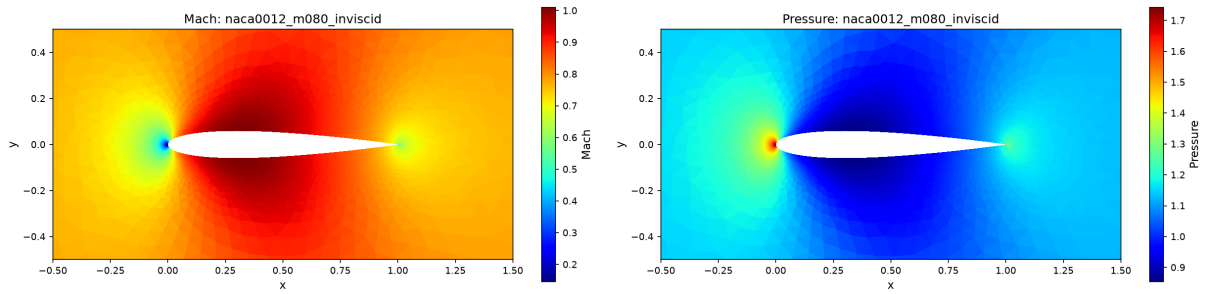


Figure 4: Mach and pressure contours (near-body): NACA 0012, $M = 0.8$, inviscid.

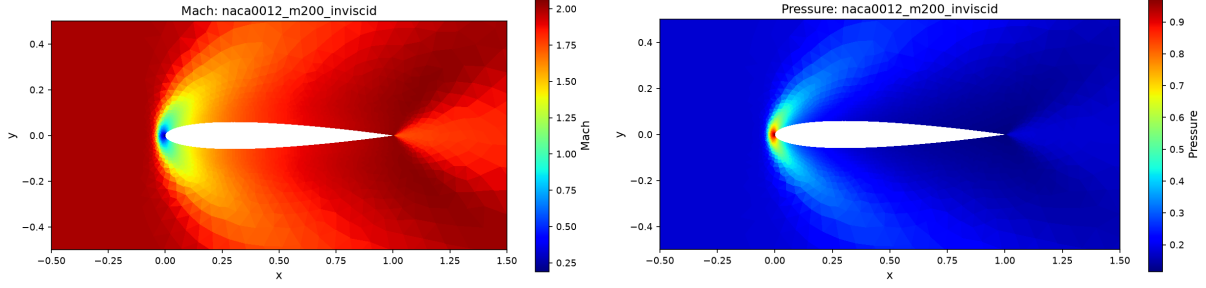


Figure 5: Mach and pressure contours (near-body): NACA 0012, $M = 2.0$, inviscid.

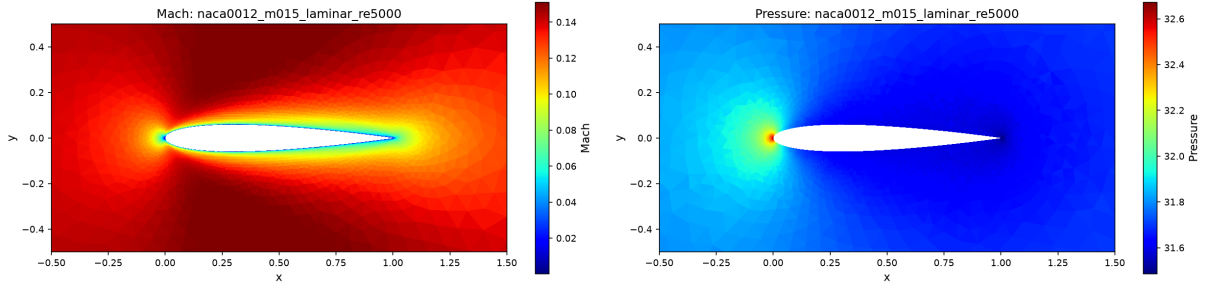


Figure 6: Mach and pressure contours (near-body): NACA 0012, $M = 0.15$, $\text{Re } 5000$.

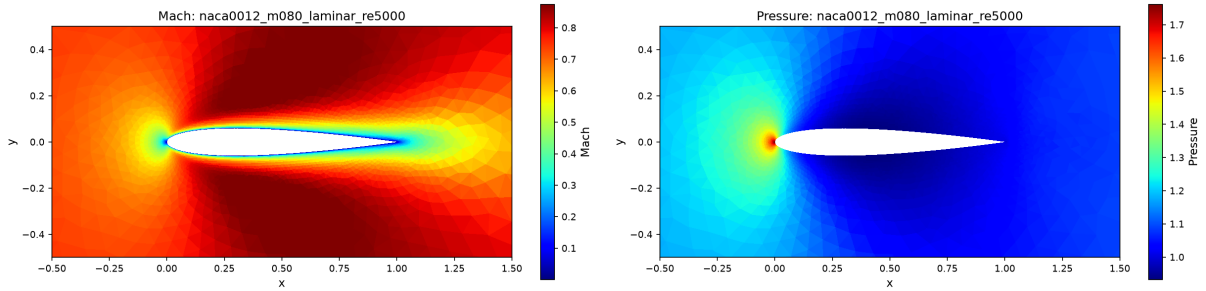


Figure 7: Mach and pressure contours (near-body): NACA 0012, $M = 0.8$, $\text{Re } 5000$.

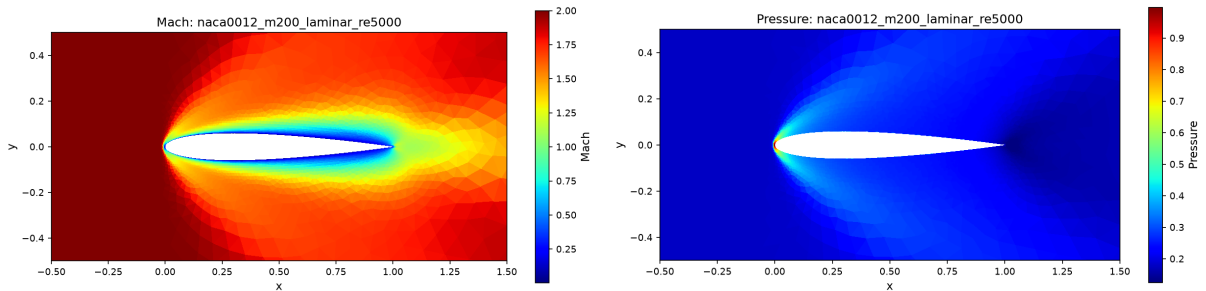


Figure 8: Mach and pressure contours (near-body): NACA 0012, $M = 2.0$, $\text{Re } 5000$.

9.3 Cylinder Reynolds 20

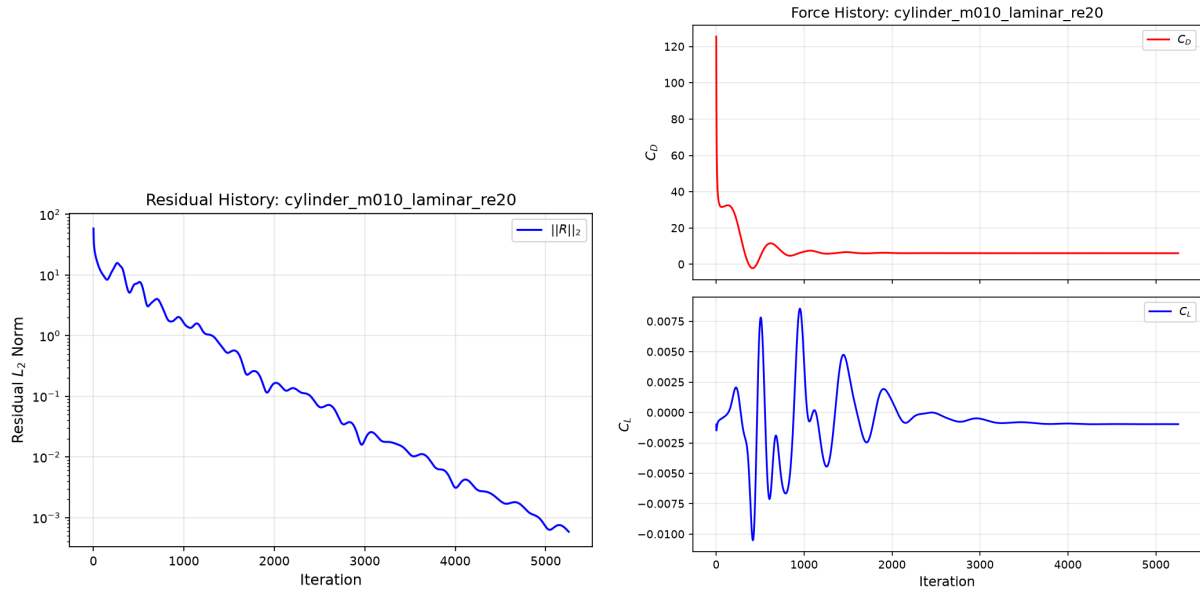


Figure 9: Residual and force history for cylinder, Re 20.

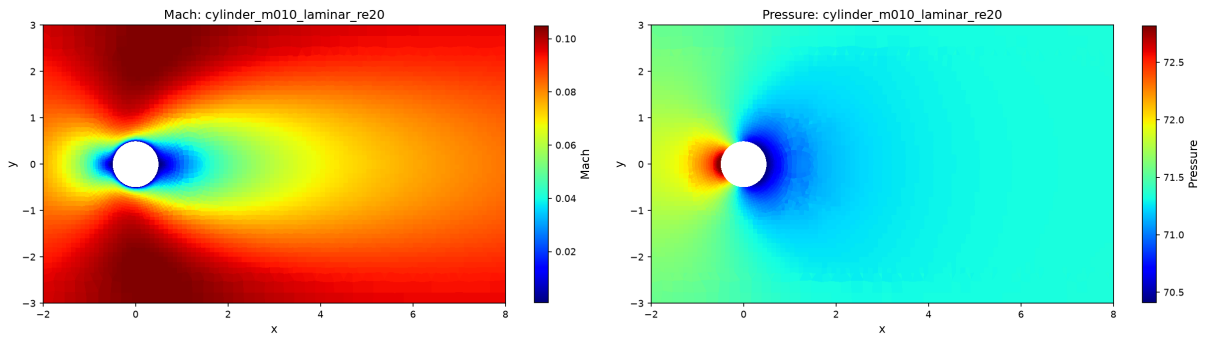


Figure 10: Mach and pressure contours (near-body): cylinder, Re 20.

9.4 Cylinder Reynolds 200 Vortex Street

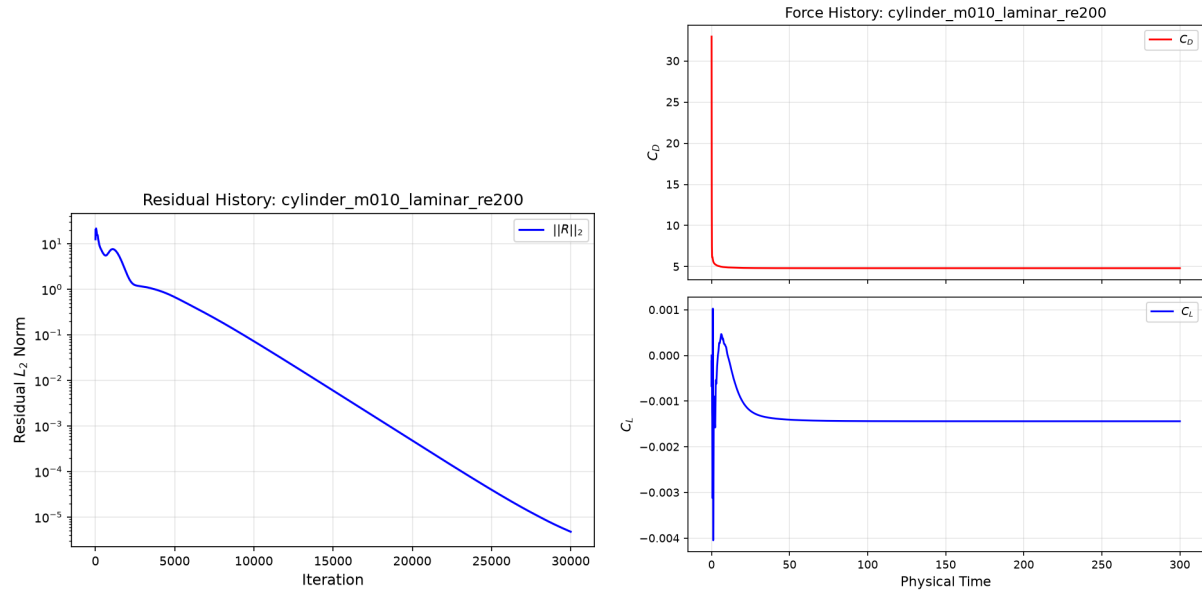


Figure 11: Residual and force history for cylinder, Re 200.

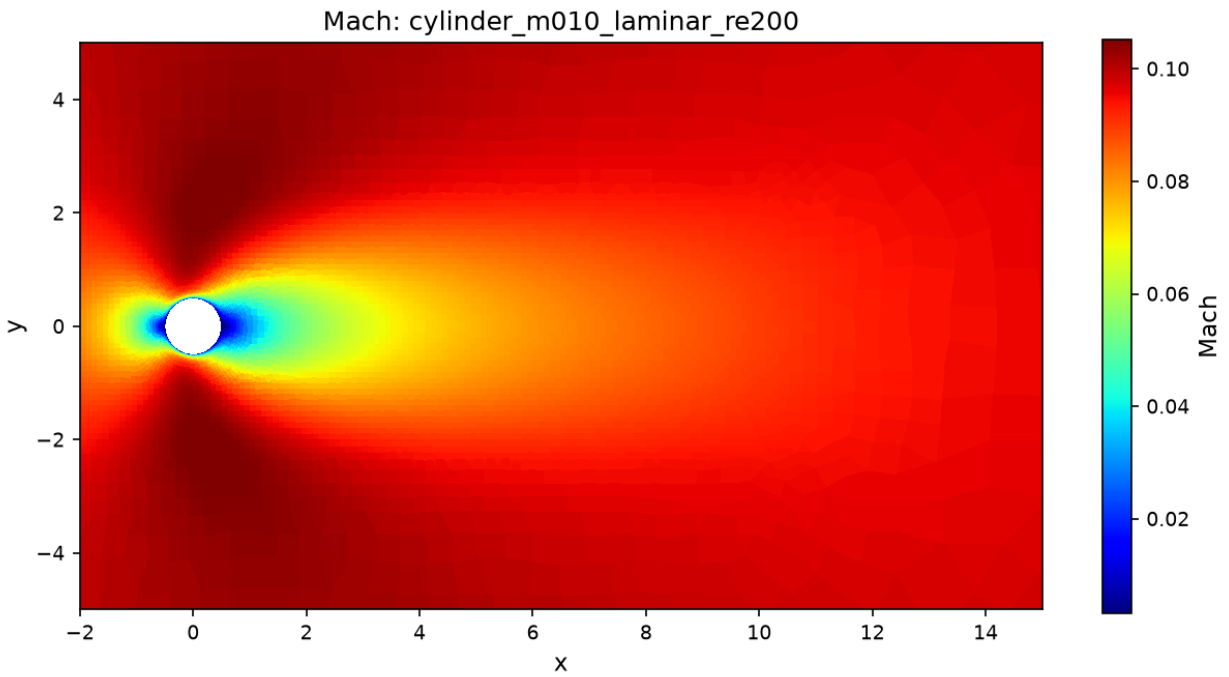


Figure 12: Mach contour (wake region): cylinder, Re 200 vortex street.

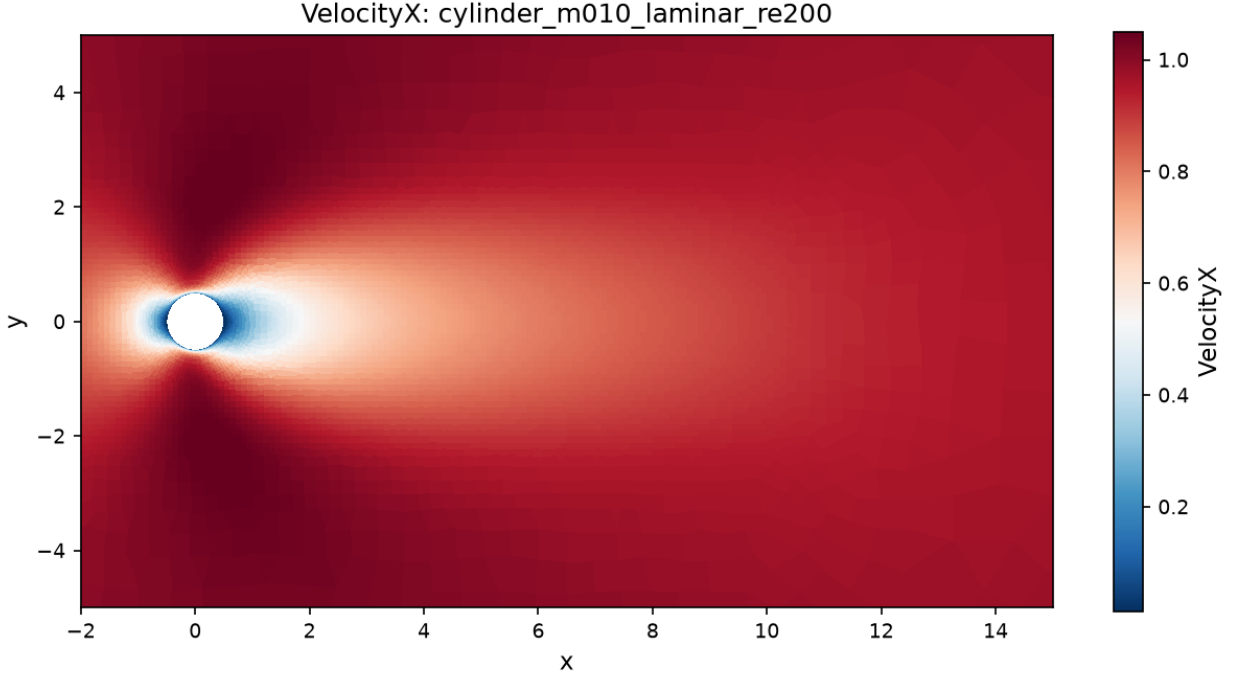


Figure 13: Velocity- x contour (wake region): cylinder, Re 200 vortex street.

10 Parallel Validation

Rank-count comparison tests were run for NACA 0012 $M = 0.15$ inviscid and cylinder Re 20 at $n_p = 4$ and $n_p = 8$.

11 Limitations and Failure Analysis

- The Barth–Jespersen limiter provides minimum acceptable oscillation control; Venkatakrishnan or MLP limiting would improve smooth-region accuracy.
- The LU-SGS implicit solver uses a simplified scalar spectral-radius Jacobian approximation. A block or matrix-free approach would improve convergence rates.
- The viscous gradient correction uses a simple direct correction along the cell-to-cell direction; a diamond-path or iterative correction would be more accurate on highly skewed meshes.